

Flying Names: Manipulating Name Tags in XR Social Environments for Target Selection Tasks

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ABSTRACT

Recent developments in Extended Reality (XR) have led to the creation of various Virtual Reality (VR) social applications that allow small groups of people to interact and participate in various activities. However, these applications are not suitable for larger-scale social scenarios. One major challenge is the lacking of effective solutions for target selection because existing methods for occluded-target selection do not apply well to humans. In this paper, we propose a method called “Flying Names” to improve decision-making accuracy during target selection for both VR and Augmented Reality (AR) social environments by manipulating the heights of name tags on the fly and connecting an auxiliary line from a name tag to its highlighted owner. Our user study results indicate that this proposal effectively improves the accuracy of identifying the actual target person and is better than using either one or none of manipulating heights and connecting lines.

CCS CONCEPTS

• Human-centered computing → User interface design.

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1 INTRODUCTION

The market for consumer-grade Virtual Reality (VR) headsets has grown significantly in recent years, leading to widespread interest in the concept of the Metaverse. The success of Meta Quest 2 has helped build a solid user base for a VR software ecosystem and the development of various social platforms and applications that provide multi-user VR experiences, including games, meetings, and movies. It is not rare that people want to communicate with specific individuals like whispering to a friend when other people are nearby. However, these platforms are currently limited to basic global-level interactions and lack fine-grained target selection mechanisms for individual-level interaction except for muting/banning/friending

another user. In contrast to VR social applications, AR social applications are less developed, with no instant communication application yet available. However, fine-grained target selection mechanisms in AR can be particularly interesting due to the added constraints imposed by the real world, which present unique challenges for design.

We believe that VR and AR wearable devices have the potential to become more seamlessly integrated into our daily lives than smartphones, functioning as external organs. However, the first step of social interactions, selecting a person, is challenging due to mutual occlusion in crowds. Target selection is a classic task that dates back to the 1980s when Foley et al. [10] defined “Select” as one of the six types of interactions for 2D graphical user interfaces (GUIs). In 3D space, there are two main categories of single-target selection methods: ray-based methods [3, 11, 16], which use a ray to intersect with objects, and cursor-based methods [12, 13, 16], which provide a 3D cursor whose location is related to the position of the user’s hand. It is possible for a method [2] to belong to both categories.

When it comes to occluded-target selection, there are two related important functionalities: visualizing the objects in a specific area and disambiguating which object is desired. Visualization techniques include enabling the user to view the environment from other directions [8, 14] and making obstructions transparent [9, 18]. To disambiguate which object is desired, techniques such as scaling [6], transforming [7], or distorting [4] the entire space or the objects can be used to make them easily distinguishable by reducing the overlap between objects.

These methods are generally designed for non-human objects and are not well-suited to AR scenarios. While manipulating non-human objects which has no free will are easy when there is only one human user, this is not the case when multiple human users are involved since operations on other humans cannot be performed without consent, for example, moving a human to another position. Furthermore, humans and objects in AR scenarios are anchored to the real world and synchronizations are almost single-sided, from the real world to the virtual space. So most operations performed in the virtual space cannot be synchronized to the real world, like distorting the 3D space. However, in XR social scenarios, the use of name tags is feasible since they are purely virtual objects that never exist in the real world and have no free will. So we can utilize name tags to help occluded-human selection. Name tags are a kind of label and researchers have studied automatic label placement algorithms, including 2D labels [5, 15] and 3D labels [1]. There are also efforts to make label placement more user-friendly by using a learning-based system [17]. However, these methods focus on

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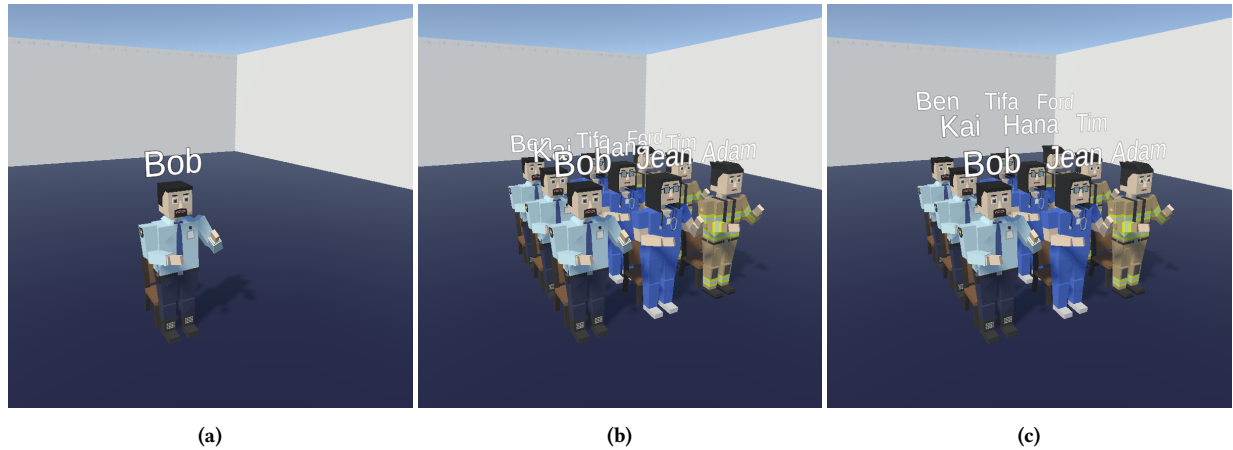


Figure 1: An illustration of our basic idea. (a) Levitating name tags above the head is a common practice. (b) Mutual occlusion occurs when people gather. (c) Our basic idea is to adjust the heights of name tags.

illustrating objects in 2D or 3D rather than helping decision-making during target selection.

In this paper, we propose a method called “Flying Names”. The core idea allows users to easily and intuitively make decisions before selecting the person they want to interact with by manipulating the height of name tags on the fly and connecting an auxiliary line from a name tag to its highlighted owner. We conducted a user study to evaluate the effectiveness of the proposed method by comparing four strategies (Baseline, ConnectLine, HighlightLowest, Flying-Names). The results show the benefit of our proposed method in the effectiveness of helping decision-making. We believe that our method has shown significant insights into the advancement of VR and AR social applications, particularly in the areas of instant communication and fine-grained interactions from global to individual levels.

2 FLYING NAMES

Our proposed method is called “Flying Names” because the name tags of people can appear to fly up and down, providing additional information to improve decision-making accuracy during target selection in crowded XR social environments.

2.1 Basic Idea and Assumptions

In 3D scenarios, it is common to display the name of a person above their head, as shown in Figure 1a. However, mutual occlusions can occur when many people are gathered closely together, as depicted in Figure 1b. In these situations, it is difficult for users to distinguish and recognize not only the people but also their names since name tags are fixed above their heads. To address this issue, it is important to consider how to visualize and disambiguate targets. Previous research has shown that there are three common approaches to avoiding visual overlap: 1) changing the viewpoint [8, 14], 2) making obstructions transparent [9, 18], and 3) transforming/distorting the whole space or objects [4, 6, 7]. However, these methods may not be directly applicable to XR social scenarios, particularly AR social scenarios.

For clarification, in XR social scenarios, the objects we want to choose are humans, and the obstructions are either humans or items like tables/chairs. During social events, people may not be able to move around arbitrarily just for a better viewpoint. In AR scenarios, the objects and the space are anchored to the real world, making it difficult to synthesize content for making obstructions transparent or transforming/distorting real-world objects. In VR scenarios, the objects and the environment are not anchored to the real world, making it possible to perform such operations. However, if we aim to find a unified solution, this introduces inconsistency, as people may participate in both VR and AR activities. It can also lead to discordance in hybrid social events that involve both VR and AR attendees.

Name tags that appear to levitate above people’s heads are virtual objects that neither exist in the real world nor be constrained by the real world. Moreover, the name tag of a certain person can appear differently from the perspective of different people. We can utilize them to help people disambiguate occluded people. Our basic idea behind the Flying Names system is straightforward: the relative height of name tags is not fixed. It can be adjusted when needed, as shown in Figure 1c. There is a primary rule that must be followed when adjusting the height of name tags: the name tag of a person should be levitating directly above the head rather than at an angle. In order to ensure that this is the case, we need to make a few basic assumptions about how people (and their heads) are located in the 3D space. Our assumptions are as follows:

- (1) People are all on a near-flat surface that is almost horizontal.
- (2) People are in a natural state that their heads have a higher altitude compared to the torsos.
- (3) People are not fully-occluded by other structures, walls for example, in the environment.
- (4) People are of similar height in a general social scenario.
- (5) People stay still during the selection process.

Among these assumptions, the first, second, and third are typically satisfied during most social events. We will not discuss how to handle situations where the fourth and fifth assumptions are not satisfied, as they are topics for future improvements.

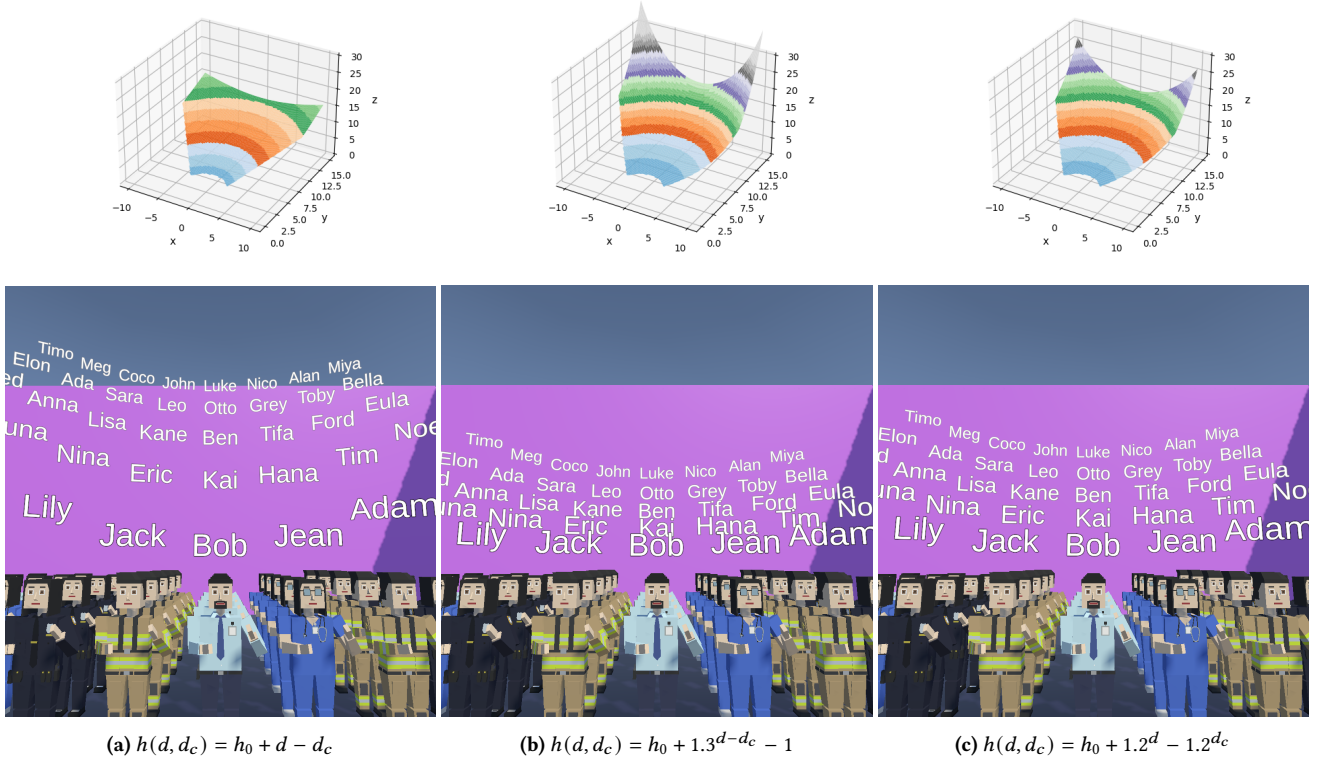


Figure 2: An illustration of the three simple parametric functions. The upper part shows the 3D visualization of the height function assuming the user is at $(0, 0, 0)$ and the closest person is at $(0, 4.3, 0)$. The lower part shows the visual effect when applied to a group of people in a grid layout. The variable h_0 represents the default height if no adjustment rules are applied, d represents the distance from a person to the user, and d_c represents the distance from the closest person to the user.

2.2 Side Effect and Solutions

In the real world, name tags or badges are attached to people’s clothes on the chest or worn around the neck, allowing others to easily recognize a person with a single glance. It is not possible for a person’s identity to be recognized by observing a name tag attached to another person. In virtual 3D worlds, name tags are often displayed slightly higher above their owners’ heads. However, when the height of the name tags is adjusted, it becomes more difficult for people to correctly match the name tag with its owner. When two people stand close together and their name tags are both very high, it is easy to make mistakes. So our basic idea introduces a side effect.

So why does this happen? It is because when there is interference, people do not have the ability to determine whether the person and the name tag have the same 2D coordinate but only differ in height. It is worth noting that since we cannot change the position of each person (because they have the freedom to decide where to be), we are improving upon occluded-target selection methods that do not involve transform operations on objects. These methods all have a disambiguation process in which the user switches between several candidates and makes a decision about each one, focusing

on one candidate at a time. If we denote the person being focused on as the “highlighted” person, then two potential solutions are:

- (1) If and only if a person is highlighted, there is an auxiliary line between the name tag and his/her head.
- (2) If and only if a person is highlighted, the name tag of him/her is placed at the normal height.

2.3 Adjustment Rules for Height of Name Tags

First, the user needs to have an overall understanding of who is in the area. We designed a light cone similar to a torch to allow users to specify an interested area. The name tags of all the people in this area will be displayed. If the user can’t find the desired target in this area, another area can be specified by waving the torch. The overlap among different name tags should be minimized so the user can see everyone’s name clearly. We decided to keep the size of name tags consistent regardless of the distance between the owner and the user, to allow users to roughly estimate the distance and avoid confusion if the name tags of two people at different distances appear the same size.

The height of a name tag can be described as a function h of the distance d from that person to the user. In order to make it easier to implement, we denoted it as $h(d, d_c)$, where d_c is the distance from the closest person in the crowd to the user, and it can be used

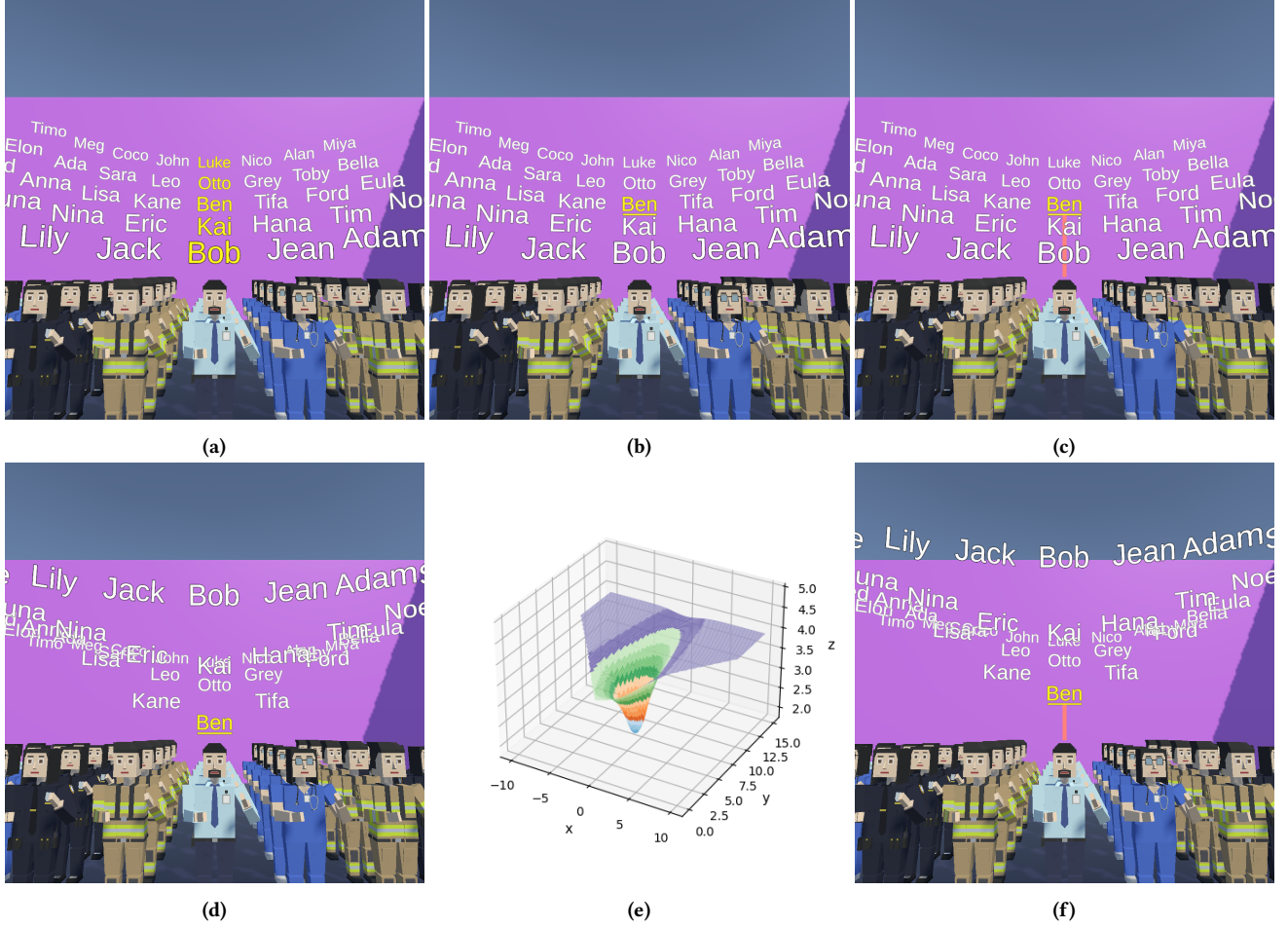


Figure 3: An illustration of disambiguation process and 3 strategies for improving target-selection performance. (a) 5 candidates with yellow name tags are intersect with the ray. (b) The baseline when no strategy is given. (c) ConnectLine strategy connects an auxiliary line as additional information. (d) HighlightLowest strategy ensures the highlighted person has the lowest name tag. (e) The corresponding 3D visualization of the height function used in HighlightLowest. (f) Both ConnectLine and HighlightLowest are used at the same time.

to satisfy the border condition. We use h_0 to represent the default fixed height of name tag when no adjustment rules are applied. There are several simple parametric functions that can produce good visual effects while still being easy to implement, such as:

- (1) $h(d, d_c) = h_0 + K(d - d_c)$, as shown in Figure 2a when $K = 1$.
- (2) $h(d, d_c) = h_0 + K^{d-d_c} - 1$, as shown in Figure 2b when $K = 1.3$.
- (3) $h(d, d_c) = h_0 + K^d - K^{d_c}$, as shown in Figure 2c when $K = 1.2$.

Figure 2 shows a crowd of people in a grid layout for the purpose of observing the visual effects of these functions more easily. In real scenarios, the implemented system can select one of these functions based on the distribution of people, and an automatic rating system can be designed to determine which function produces the best visual effects. Currently, we have not designed such a rating system, so we are using the third function, referred to as the “baseline”, directly. This height function is set as the baseline instead of everyone’s name tag at the default height h_0 because

all the name tags will cause severe mutual occlusion too if their heights are the same, which is impossible for users to observe the name of each person.

After the user finds the target person in the area and uses a ray to intersect with several candidates, as shown in Figure 3a (for illustration purposes, all the people in the room are still in a perfect grid layout). If no additional information is provided, the user must switch between these 5 candidates. Figure 3b shows an example in which Ben becomes the highlighted person. As previously described, we have two strategies to help the user determine whether the highlighted person is the target:

- **ConnectLine:** We can add an auxiliary line from Ben’s name tag to his head as shown in Figure 3c.
- **HighlightLowest:** We can ensure Ben has the lowest name tag as shown in Figure 3d.
- **FlyingNames:** Our final proposal is using both ConnectLine and HighlightLowest as shown in Figure 3f.

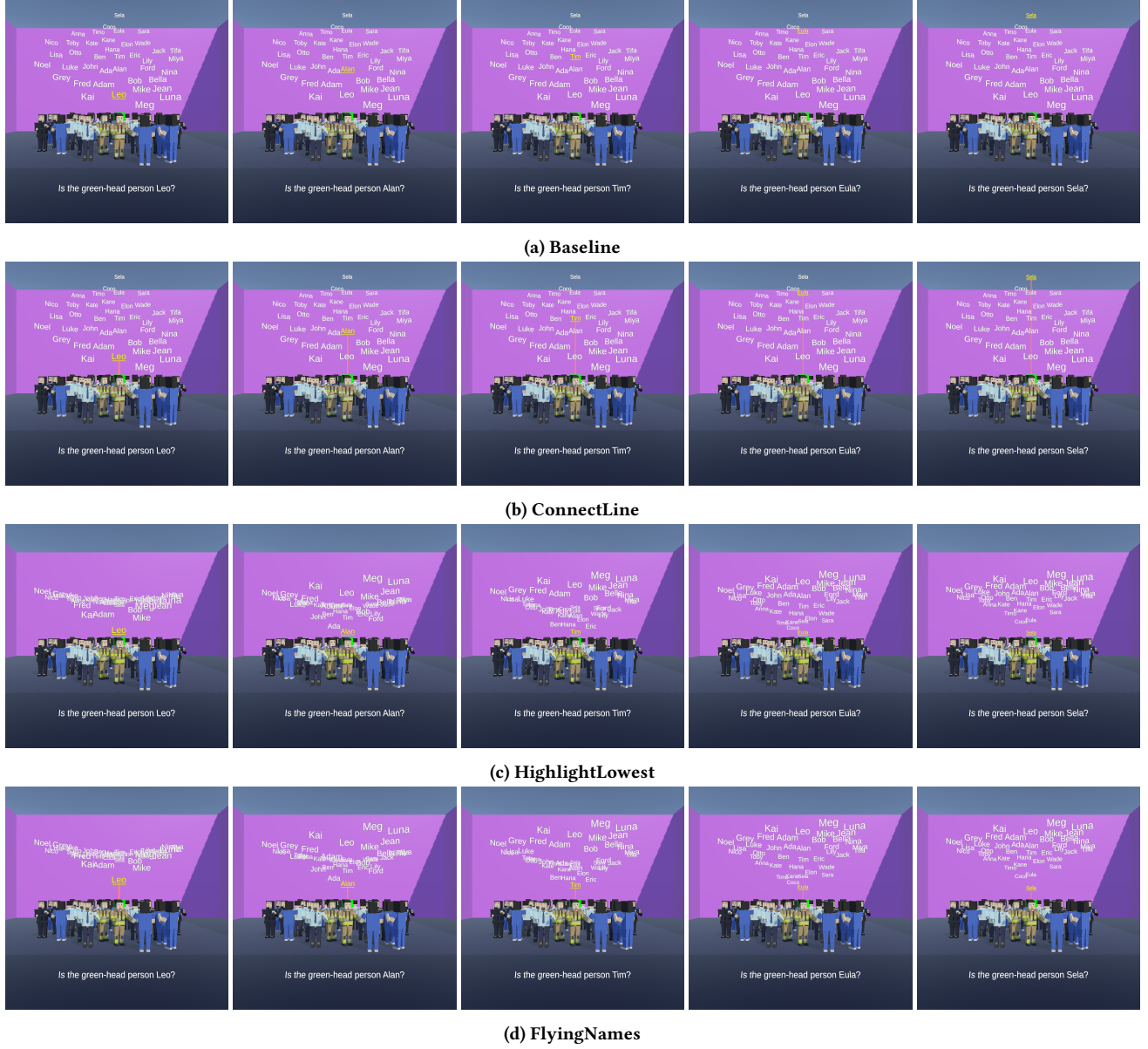


Figure 4: An example of the 20 different first type of questions from the same scenario with different settings and highlighted candidate. The visual effect of second type of questions are just animated version of these first type of questions.

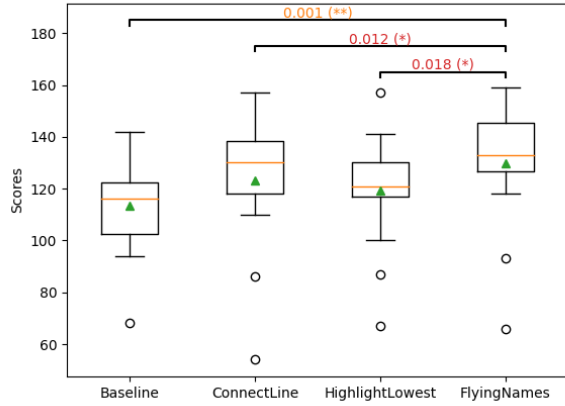
Let P_o , P_u , and P_h represent the positions of the owner of the name tag, the user, and the highlighted candidate respectively, then we have

$$\begin{aligned} d_1 &= \|P_o - P_u\| \\ d_2 &= \left\| \frac{P_h - P_u}{\|P_h - P_u\|} \times (P_o - P_u) \right\| \\ d_3 &= \|P_h - P_u\| \end{aligned}$$

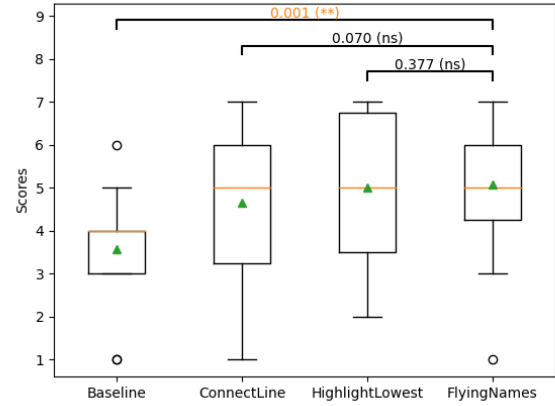
Using these three variables, we designed a funnel-shaped height function:

$$h(d_1, d_2, d_3) = h_0 - K_1^{-d_2^2} \cdot K_2^{2-|d_1-d_3|} + K_2^2.$$

Figure 3e shows the 3D illustration when $P_u = (0, 0, 0)$, $P_h = (0, 6.3, 0)$, $K_1 = 1.4$, and $K_2 = 1.2$. This function ensures that only the highlighted person has the lowest name tag and can be seen clearly. The names of people on the left and right can also be easily recognized because the function uses d_2 , which is the projected distance between the name tag owner and the highlighted candidate. This helps to minimize overlap among the name tags.



(a) First type of questions: Is the green-head person (Bob)?



(b) Second type of questions: Who is the green-head person?

Figure 5: The Boxplots for the scores of the the two types of questions. The P-values calculated from the Wilcoxon signed-rank tests are shown as annotations.

3 USER STUDY

We conducted a user study on our proposal, which also served as an ablation study to evaluate the effectiveness of ConnectLine and HighlightLowest.

3.1 Participants, Apparatus, and Materials

We recruited 14 university students (9 male, 5 female) aged between 20-26 with diverse educational backgrounds. All participants had a normal or correct-to-normal vision. Because the focus of our user study was on the effectiveness of each strategy in improving judgment accuracy during the disambiguation process, they were only required to wear a Meta Quest 2 for observing the scenarios and answer multiple-choice questions using a Meta Quest 2 Right Controller. Previous experience with VR/AR devices was not required for participation in our user study, as participants were not required to complete tasks such as actually selecting targets and being timed.

3.2 Design and Procedure

Our user study employed a within-subjects design, wherein each participant evaluated all four label arrangement systems (Baseline, ConnectLine, HighlightLowest, FlyingNames) across seven similar but distinct scenarios. Each scenario consisted of a dense crowd of 40 people, including a predefined target person with a green head and five predefined candidates (including the green-headed person) arranged along a ray. Different scenarios have different distribution of the 40 people. The questions posed in this form were of two types:

- (1) “Is the green-head person Bob?” where participants judged whether the green-head person corresponded to the name tag that showed Bob. The choices are “Yes”, “Maybe Yes”,

“Maybe No”, and “No”. If the participants are not very confident about choosing “Yes” or “No”, they can choose “Maybe Yes” or “Maybe No” instead.

- (2) “Who is the green-head person?”, in which participants chose from the 5 given candidate names. The name tags were animated, automatically switching the highlighted person among the 5 candidates for observation by the participants. The purpose of designing this type of questions is to evaluate how the switching animation impacts user experience, while minimizing the influence of participants’ individual switching behavior on the results.

Each scenario has 20 questions for the first type (5 questions for each arrangement system showing 5 different highlighted candidates) and 4 questions for the second type (1 question for each arrangement system). Figure 4 shows one of the scenarios as an example. These questions are designed to be very difficult, which allows for good variation in the correctness of each arrangement system for comparison. A total of 168 questions were randomly shuffled in advance to ensure that questions from the same scenario were not grouped together, preventing participants from relying on vague memory of the previous question to answer the current one.

We first introduced the aim of our user study and explained how the scenarios would look like and how to answer questions using the controller. Participants then had to go through several tutorial questions before answering the actual questions. The entire process took approximately 20-30 minutes. They were compensated with a 600 JPY Amazon gift card after the user study.

3.3 Results

For the first type of task, the correct answer could only be “Yes” or “No”, so we used the following criteria to score participants’ answers: 5 points were awarded for a correct answer with high confidence (i.e., “Yes” or “No”), 3 points were awarded for a correct answer with low confidence (i.e., “Maybe Yes” or “Maybe No”), 1

point was awarded for an incorrect answer with low confidence, and 0 points were awarded for a totally incorrect answer. For the second type of task, 1 point was awarded for a correct answer and 0 points were awarded for an incorrect answer.

We performed Wilcoxon signed-rank tests using SciPy on the scores due to the small sample size. The results presented in Figure 5a indicate that, for the first type of task, FlyingNames outperformed Baseline significantly (p -value=0.001, effect size=0.764), ConnectLine significantly (p -value=0.012, effect size=0.596), and HighlightLowest significantly (p -value=0.018, effect size=0.562). In Figure 5b, for the second type of task, FlyingNames showed significant improvement over Baseline (p -value=0.001, effect size=0.796). However, there was no significant difference between FlyingNames and ConnectLine (p -value=0.070, effect size=0.394), and between FlyingNames and HighlightLowest (p -value=0.377, effect size=0.083). One possible explanation is that using a 5-choice question implies that the correct answer is among the choices, making it easier to guess, whereas in real scenarios, it is not guaranteed that the desired person is actually on the ray, as it is difficult to use the ray to intersect with a person in a crowd.

4 CONCLUSION, DISCUSSION AND FUTURE WORK

We proposed a novel and effective method called “Flying Names” that manipulates the name tags of people on the fly to help users decide whether a candidate is the actual desired target. Its core concept is that by manipulating the height of name tags, the mutual occlusion of crowds can be reduced, allowing people to clearly observe all name tags in the virtual world instead of observing the crowds directly. It consists of two strategies called **ConnectLine** and **HighlightLowest** which are used to minimize an introduced side effect that people may lose the ability to perceive the correspondence between name tags and their owners. ConnectLine connects an auxiliary line from a name tag to its highlighted owner, while HighlightLowest ensures the highlighted person has the lowest name tag. Our user study results showed that our proposal significantly improves the accuracy of deciding whether a highlighted person is the actual desired target for selection and using these two strategies together has a synergistic effect compared to using only one of them alone. However, in cases where the desired target is guaranteed to be among several candidates, which is a topic of our future research, using both strategies together only marginally improves accuracy compared to using either strategy alone due to the decrease in difficulty.

Our current proposal primarily aims to assist users in connecting a person’s appearance with their name. However, the approach can also be used for selecting distinguishable objects other than humans. While previous methods [4, 6–9, 14, 18] can be easily applied in VR spaces, our method offers advantages in AR spaces because it does not attempt to affect real-world objects. As Internet-of-Things technology becomes more prevalent, we will have the ability to check the names of objects in the real world, enabling the use of selection by name tags. This advancement will make it possible to extend the application of our approach beyond humans to other objects, further increasing its utility.

The next step in our research is to improve the feedback mechanism. This will allow the user to confirm that the desired person is being intersected by the ray and avoid any unnecessary decision-making. This is a crucial improvement that will help make our target selection method fully functional. In addition to the potential improvement in the feedback mechanism, another important aspect of future research is to remove the assumption that people are of similar heights and they do not move around. There is a significant unexplored area in developing fast and accurate moving target selection when occlusion occurs regularly, as these two requirements can conflict with each other. This area has received little attention, but it is essential for building seamless XR social experiences, particularly in AR scenarios. Luckily, it is very rare for a moving target to be obscured by another object for a long time, so there are only static targets being occluded by moving obstruction or moving targets being occluded by static obstruction. For these two cases, occlusion is just a short-term interference.

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REFERENCES

- [1] Blaine Bell, Steven Feiner, and Tobias Höllerer. 2001. View management for virtual and augmented reality. In *Proceedings of the 14th annual ACM symposium on User interface software and technology*. 101–110.
- [2] Hrvoje Benko and Steven Feiner. 2007. Balloon selection: A multi-finger technique for accurate low-fatigue 3d selection. In *2007 IEEE symposium on 3D user interfaces*. IEEE.
- [3] Richard A Bolt. 1980. “Put-that-there” Voice and gesture at the graphics interface. In *Proceedings of the 7th annual conference on Computer graphics and interactive techniques*. 262–270.
- [4] M Sheelagh T Carpendale, David J Cowperthwaite, and F David Fracchia. 1996. Distortion viewing techniques for 3-dimensional data. In *Proceedings IEEE Symposium on Information Visualization '96*. IEEE, 46–53.
- [5] Jon Christensen, Joe Marks, and Stuart Shieber. 1995. An empirical study of algorithms for point-feature label placement. *ACM Transactions on Graphics (TOG)* 14, 3 (1995), 203–232.
- [6] Mei C Chuah, Steven F Roth, Joe Mattis, and John Kolojechick. 1995. SDM: Selective dynamic manipulation of visualizations. In *Proceedings of the 8th annual ACM symposium on User interface and software technology*. 61–70.
- [7] Niklas Elmqvist. 2005. BalloonProbe: Reducing occlusion in 3D using interactive space distortion. In *Proceedings of the ACM symposium on Virtual reality software and technology*. 134–137.
- [8] T Todd Elvins, David R Nadeau, and David Kirsh. 1997. Worldlets—3D thumbnails for wayfinding in virtual environments. In *Proceedings of the 10th annual ACM symposium on User interface software and technology*. 21–30.
- [9] Steven K Feiner and Dorée Duncan Seligmann. 1992. Cutaways and ghosting: satisfying visibility constraints in dynamic 3D illustrations. *The Visual Computer* 8, 5 (1992), 292–302.
- [10] James D Foley, Victor L Wallace, and Peggy Chan. 1984. The human factors of computer graphics interaction techniques. *IEEE computer Graphics and Applications* 4, 11 (1984), 13–48.
- [11] Andrew Forsberg, Kenneth Herndon, and Robert Zeleznik. 1996. Aperture based selection for immersive virtual environments. In *Proceedings of the 9th annual ACM symposium on User interface software and technology*. 95–96.
- [12] Mark R Mine, Frederick P Brooks Jr, and Carlo H Sequin. 1997. Moving objects in space: exploiting proprioception in virtual-environment interaction. In *Proceedings of the 24th annual conference on Computer graphics and interactive techniques*. 19–26.
- [13] Ivan Poupyrev, Mark Billinghurst, Suzanne Weghorst, and Tadao Ichikawa. 1996. The go-go interaction technique: non-linear mapping for direct manipulation in VR. In *Proceedings of the 9th annual ACM symposium on User interface software and technology*. 79–80.
- [14] Richard Stokley, Matthew J Conway, and Randy Pausch. 1995. Virtual reality on a WIM: interactive worlds in miniature. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 265–272.

- [15] Jan W van Roessel. 1989. An algorithm for locating candidate labeling boxes within a polygon. *The American Cartographer* 16, 3 (1989), 201–209.
- [16] Donald Lee Vickers. 1972. *Sorcerer's apprentice: head-mounted display and wand*. The University of Utah.
- [17] Ian Vollick, Daniel Vogel, Maneesh Agrawala, and Aaron Hertzmann. 2007. Specifying label layout style by example. In *Proceedings of the 20th annual ACM symposium on User interface software and technology*. 221–230.
- [18] Shumin Zhai, William Buxton, and Paul Milgram. 1996. The partial-occlusion effect: Utilizing semitransparency in 3D human-computer interaction. *ACM Transactions on Computer-Human Interaction (TOCHI)* 3, 3 (1996), 254–284.